

2.1 Offline Mode

Off-line mode does system setting and self diagnostic etc. Before operating, it prepares here.

IMPORTANT

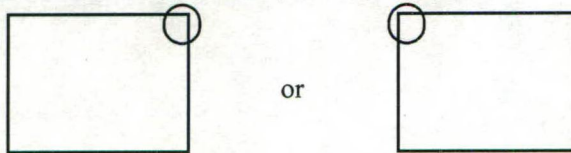
- OFFLINE mode is unavailable in a completely new GP until the necessary GP system data has been transferred from GP screen editor software. To do this, be sure the GP's power cord is plugged in and when transfer screen data from GP screen editor software to the GP, GP's system data will be automatically sent. For more information about data transfer, refer to GP-Pro EX Reference Manual.

2.1.1 Entering OFFLINE Mode

There is offline mode for various setup required at use in GP. There are two ways to enter OFFLINE mode. First, is immediately after plugging in the GP's power cord, and second, by using the Forced Reset feature.

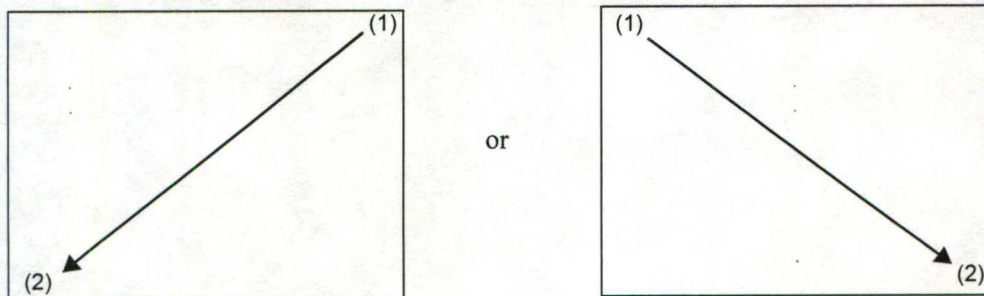
■ After Plugging in the Power Cord

Touch the upper right corner or upper left corner (within 40 pixels of the edges) of the panel for at least 3 seconds soon after the startup screen is displayed.



■ When Operating

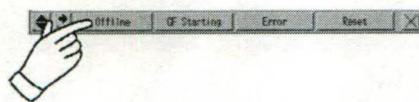
- 1 Touch either the upper right and lower left corners, or the upper left and lower right corners (within 40 pixels of the edges) of the panel in this order within 0.5 seconds.



- 2 System menu is displayed and touch "Offline" button.

For details of the buttons of the system menu, see the following section.

☞ "2.15.12 System Menu" (page 2-171)



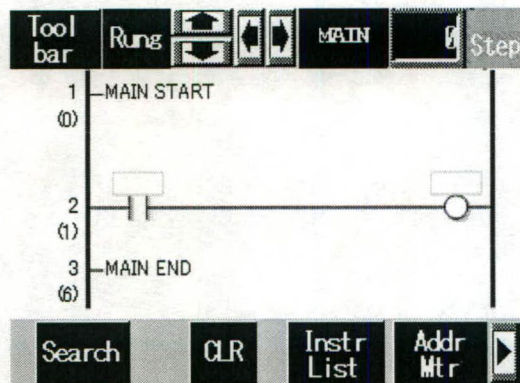
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29.12.2 Logic Monitor Functions

The following explains the logic monitor features.

■ Logic monitor

Monitors the entire logic. The logic monitor allows you to check the operational status and instruction layouts.

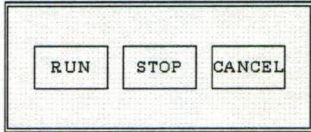


The logic monitor has the following features.

Feature		Details
Scroll		Scroll the logic using [Rung] or [Column]. Rung: Scroll the logic using rungs. Column: Scroll the instructions one by one without the logic. For landscape, you can use only the [Column] scroll.
Zoom		Touch the displayed instruction to zoom the monitor display. ☞ "■ Zoom Monitor" (page 29-94)
Logic Name Display		Display the logic names being monitored. The names to be displayed are [INIT], [MAIN], [ERRH], and [SUB-01]-[SUB-32].
Step		Display the top step number being monitored. When any change is made, the operation jumps to the rung with the specified step number.
Toolbar		Switch the toolbar show/hide at the bottom of the screen. Page 1 ↕ Page 2 Click or to switch Page 1 with Page 2.
Exit		End the monitor.

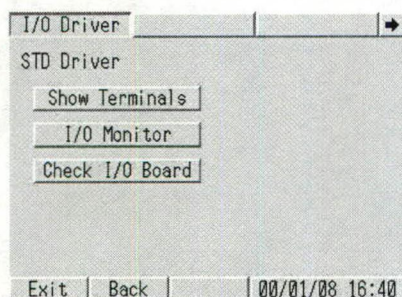
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Monitoring Logic Programs on the GP (Logic Monitor)

Feature		Details
Toolbar	RUN/STOP 	Switch the logic between RUN and STOP. Click to display the screen below. Use the buttons to run and stop the logic. 
	Address Monitor 	Switch to the address monitor.  " ■ Address Monitor" (page 29-95)
	Ladder Instructions 	Switch to the instruction list.  " ■ Ladder Instructions" (page 29-97)
	Search 	Search the variables and instructions specified in the instruction list.  " ■ Search" (page 29-98)

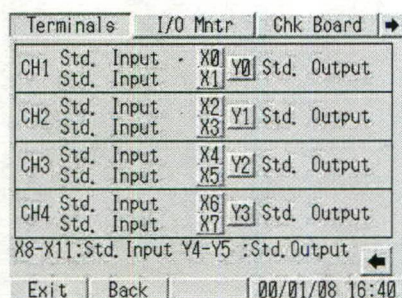
■ [Peripheral Settings] Setting Guide

◆ [I/O Driver] (STD Driver)



Setting item	Description
Show Terminals	The terminal configuration display screen is displayed.
I/O Monitor	The I/O monitor screen is displayed.
Check I/O Board	The I/O board check execution screen is displayed.

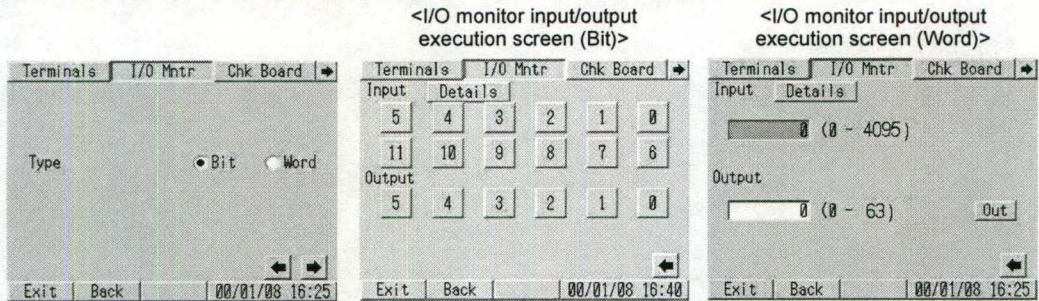
- [Show Terminals]

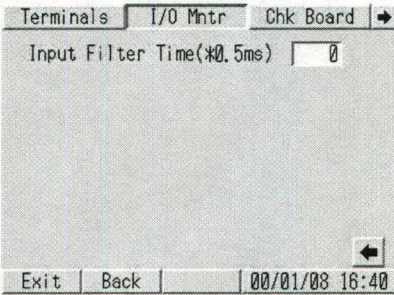
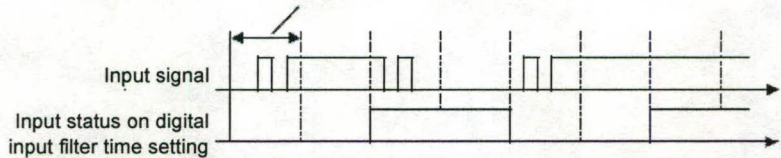


Setting item	Description
Terminals	Configurations of the specified input terminals (X0 to X11) and output terminals (Y0 to Y5) are displayed. NOTE When the LT33** series is used, it displays the configuration of input terminals X0 to X15 and output terminals Y0 to Y15. The bottom of the screen shows [X8-X15: Std. Input] and [Y4-Y15: Std. Output].

• [I/O Monitor]

The monitoring results of the standard input and output can be displayed by selecting either bit or integer. The [Word] display cannot be selected when special I/O is used.

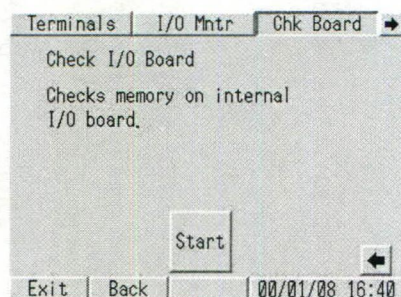


Setting item	Description
Details	The following details setting screen appears.  On the details setting screen, specify an input filter time in the range of 0 to 40. The input filter eliminates noise from an input signal. An input signal that does not reach the time duration specified here will not work. The system samples data at 0.5 ms intervals and stores the data internally, to verify the data by reading the input terminal status before the specified time at an I/O refresh interval of 2 ms. If the input terminal is always in the same state, this state is regarded as the value for the input terminal. If it is not in the same state, the previous value is regarded as the input value. Digital input filter time 

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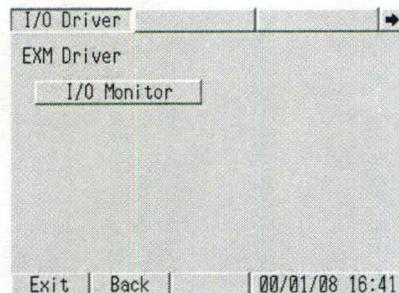
Setting item	Description
Input (0 to 11)	- When [Type] is set to [Bit] The terminal number specified for standard input is displayed. You can check the input value by bit. When the switch is not pressed, the OFF value is displayed. When the switch is pressed, the ON value is displayed. NOTE - With the LT33** series, you can check the status by using inputs 0 through 15. - When [Type] is set to [Word] You can check the input value with integers 0 to 4095 when the LT32** series is used, or with integers 0 to 65535 when the LT33** series is used.
Output (0 to 5)	- When [Type] is set to [Bit] The terminal number specified for standard output is displayed. You can check the output value by bit. NOTE - With the LT33** series, you can check the status by using outputs 0 through 15. - When [Type] is set to [Word] You can check the output value with integers 0 to 63 when the LT32** series is used, or with integers 0 to 65535 when the LT33** series is used.
Error display	If the I/O driver detects an error, the corresponding error code and error message are displayed. For details about error messages, refer to the following description. ☞ "1.7.3 Errors displayed with the LT3000 series ■ I/O driver errors" (page 1-199)

- [Check Board]



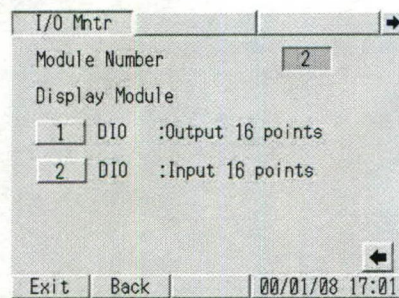
Setting item	Description
Start	The system starts checking if the I/O board is operating normally or not. If the I/O board operation is abnormal, the corresponding error message appears. For details about error messages, refer to the following description. ☞ "1.7.3 Errors displayed with the LT3000 series ■ I/O driver errors" (page 1-199)

◆ [I/O Driver] (EXM Driver)



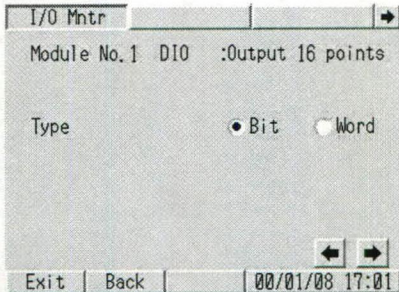
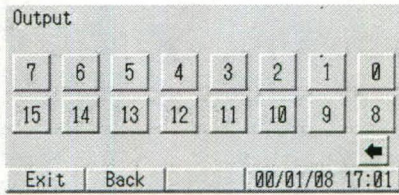
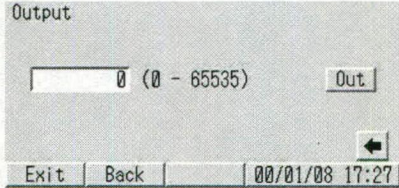
Setting item	Description
I/O Monitor	The I/O monitor connection module information screen is displayed.

• [I/O Monitor] (DIO settings)



Setting item	Description
Module Number	The number of connected modules is displayed. (0 to 2) NOTE • When the LT33** series is used, it displays the number of connected modules with values 0 to 3.
Display Module	The number of connected modules is displayed. Module No. 1 indicates the module directly connected to the LT back panel.

Continued

Setting item	Description
Module No.	The I/O monitor setting screen is displayed. I/O monitor setting screen Type 
	I/O execution screen (Bit) Outputs signals to a destination module on a bit basis. Input status after I/O monitoring appears in a bit representation when selecting the input module. 
	I/O execution screen (Word) Sets output values transmitted to a destination module and output values appear. Input values after I/O monitoring appear when selecting the input module. 

- [I/O Monitor] (Analog settings)

I/O monitor connection module information screen

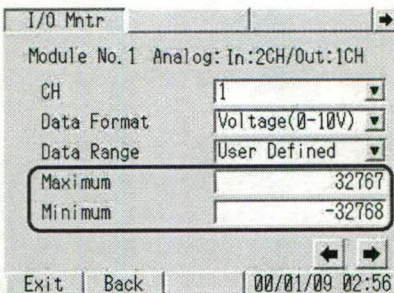
Setting item	Description
Module Number	The number of connected modules is displayed. (0 to 2) NOTE • When the LT33** series is used, it displays the number of connected modules with values 0 to 3.
Display Module	The number of connected modules is displayed. Module No. 1 indicates the module directly connected to the LT back panel.
Module No.	The I/O monitor setting screen is displayed.

I/O monitor connection module information screen

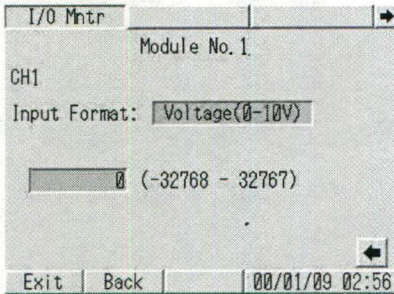
Setting item	Description
CH	Specify the number of channels for execution of the I/O monitor. NOTE • If “In: 2CH/Out: 1CH” is selected, the third channel is used for output.
Data Format	Select the data format for execution of the I/O monitor: “Voltage (0 - 10V)”, “Current (4 - 20 mA)”, “Pt100”, “K Thermocouple”, “J Thermocouple” or “T Thermocouple”.
Data Range	Select the data range for execution of the I/O monitor: “Fixed”, “Centi-grade”, “Fahrenheit” or “User Defined”.

Continued

Offline settings guide

Setting item	Description
Maximum / Minimum	Specify the upper and lower limit values for execution of the I/O monitor. These parameters can be specified only when “User Defined” is selected for “Data Range”. 

I/O monitor analog input execution screen



Setting item	Description
Input Format	Displays the “Data Format” specified on the I/O monitor setting screen.
Input value	Displays the input value.
Input data range	Displays the “Data Range” specified on the I/O monitor setting screen.

Continued

I/O monitor analog output execution screen

I/O Mnt'r

Module No. 1

CH1

Output Format: Voltage(0-10V)

0 (-32768 - 32767) Out

Exit Back 00/01/09 02:56

Setting item	Description
Output Format	Displays the “Data Format” specified on the I/O monitor setting screen.
Output value	Displays the output value. Touching the entry field displays numeric keys, allowing you to specify an output value.
Output Range	Displays the “Data Range” specified on the I/O monitor setting screen.
▲▼	Increases or decreases the output value.
Output	Outputs the value specified in “Output value”.

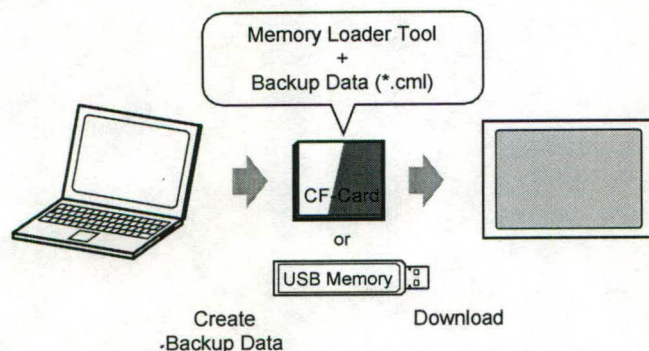
33.7 Transferring Project Files Using a CF Card or USB Storage

33.7.1 Introduction

Without connecting USB or Ethernet cables, you can transfer project files between the GP and your PC using an external memory such as a CF Card or USB storage device. You can also use the external memory to copy project files from one GP to another.

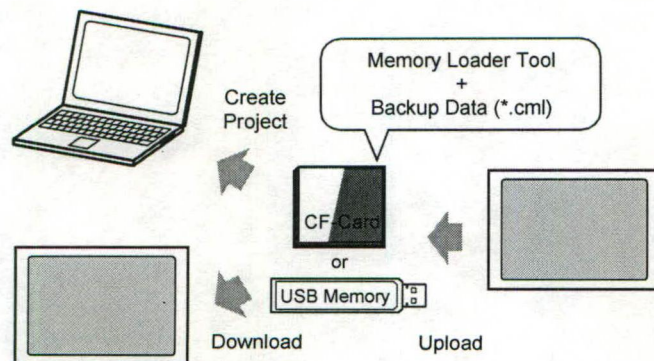
- Transfer from PC to GP

On the GP, download and displays the project backup data stored on a CF Card or USB storage device.



- Transfer from GP to PC

Uploads the backup data from the GP to a CF Card or USB storage device and loads the data to the PC as a project file. You can also download and display the backup data on a different GP.

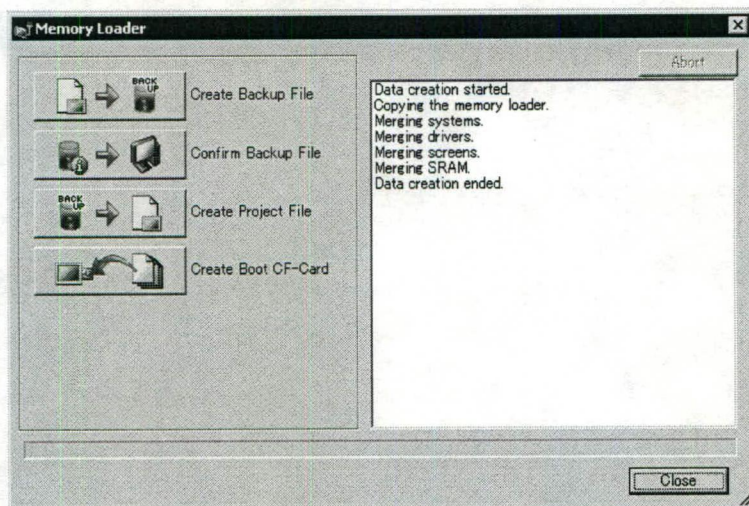


IMPORTANT

- To backup (upload) the project on the GP or start (download) the project from the backup data on a CF Card, the special Memory Loader Tool must also be copied to the CF Card
To backup the project on a USB storage device, a different Memory Loader Tool specific for USB storage devices must be copied to the USB storage device.
- Once the data is saved in the CF Card folder, you cannot use the data in USB storage. Similarly, once the data is saved in the USB destination folder, you cannot use the data on a CF Card.

Transferring Project Files Using a CF Card or USB Storage

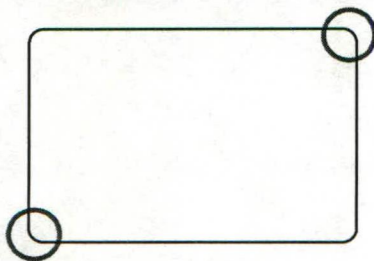
7 When the message "Data Creation Ended" appears, click [Close].



8 Copy or move all data created in the CF Card folder to the CF Card. The data, project backup data, and the CF Card Memory Loader Tool are all transferred to the CF Card.

9 Insert the created CF Card in the GP.

Touch the top right corner -> bottom left corner (or top left corner -> bottom right) on the GP screen in this order within 0.5 seconds and switch to offline mode.

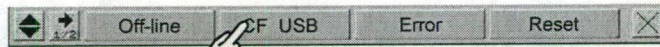


NOTE

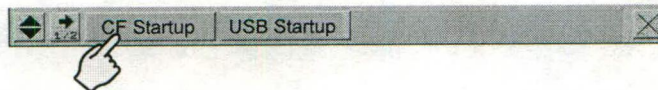
- To start the CF Card, you can either use the system menu or the DIP switches on the back of the GP. Turn the GP power OFF, turn ON DIP switch 1, and then turn the GP power back ON to launch the Memory Loader tool. You cannot use the DIP switches to start the CF Card when using USB Storage.
- To transfer project data to the GP for the first time (for a newly purchased GP), the [Initial Transfer Mode] screen appears when you turn the power ON.

Transferring Project Files Using a CF Card or USB Storage

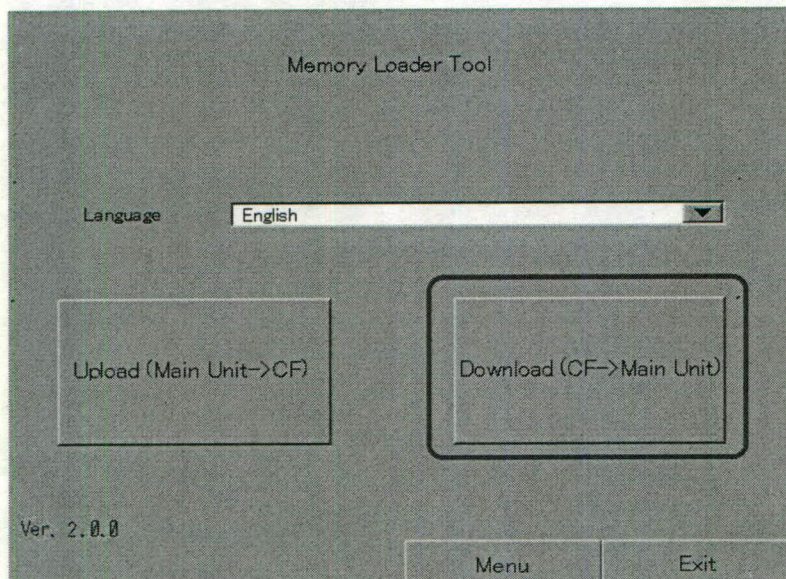
10 Touch [CF/USB]. Touch [CF startup] to reset GP.



Touch [CF/USB].

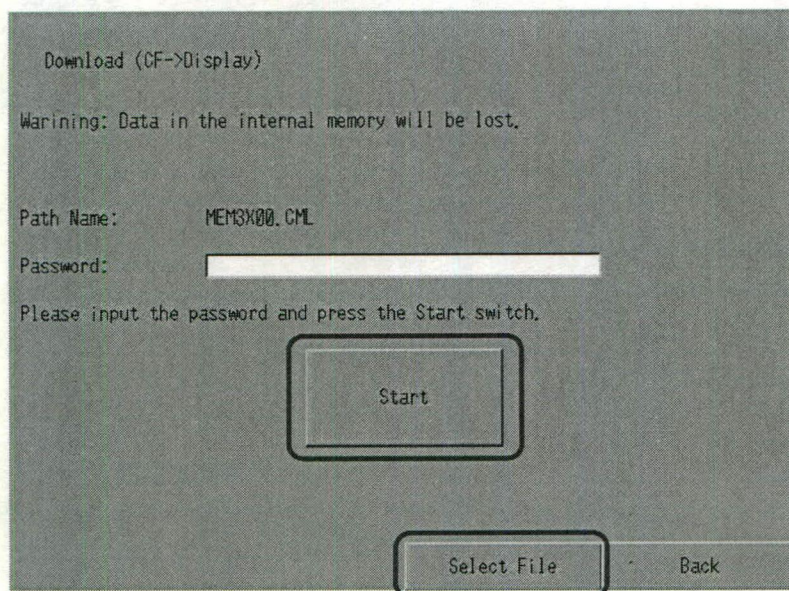


11 The Memory Loader Tool launches. The following screen opens. Touch [Download].



Transferring Project Files Using a CF Card or USB Storage

- 12 In [Select File], select a file you want to transfer and touch [Start]. If there is a password for transferring, input the password first, then touch [Start].



IMPORTANT

- Once you start downloading, all project data on the GP is deleted, including the data in the backup SRAM.
- If you performed CF Start with the DIP switch, turn the Number 1 DIP switch back OFF.

- 13 Once downloading is complete, touch [Back] and then [Exit] to exit the Memory Loader Tool.

Transferring Project Files Using a CF Card or USB Storage

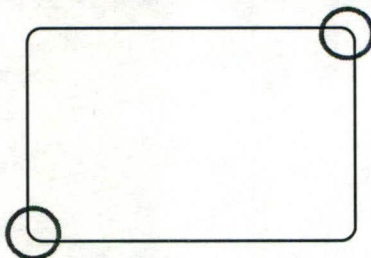
- 4 Move or copy all the created data to the CF Card/USB storage. The CF Card/USB storage device is now ready.

◆ Transfer (GP → CF Card → PC)

The following shows the process for using a CF Card. To use USB Storage, replace "CF Card" with "USB storage" in the process.

- 1 Insert the CF card with boot data.

To display the system menu, within 0.5 seconds of each touch, touch the top-right then the bottom-left corner, or touch the top-left then bottom-right corner.



- 2 Touch [CF/USB]. Touch [CF startup] to reset GP.

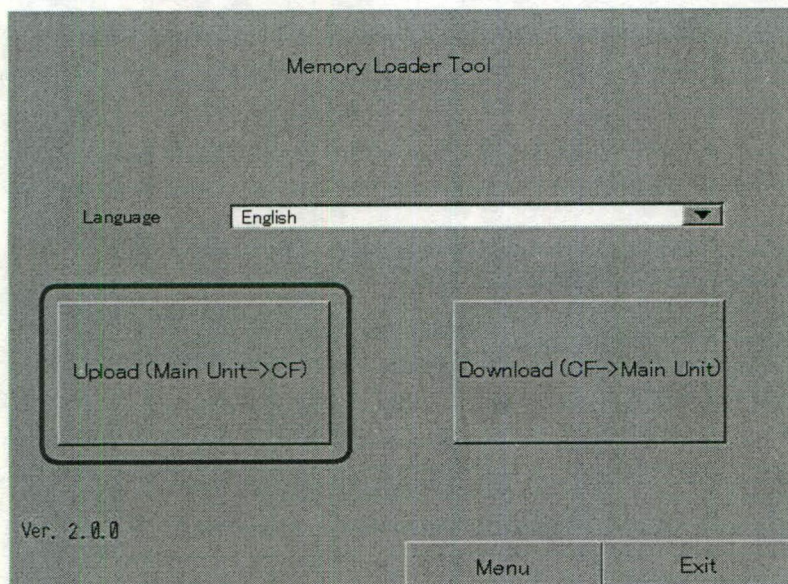


Touch [CF/USB].

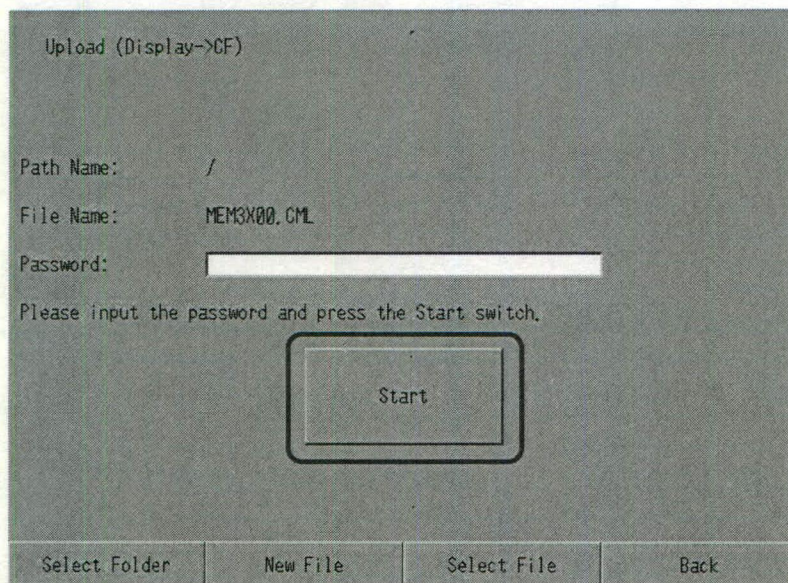


Transferring Project Files Using a CF Card or USB Storage

- 3 The Memory Loader Tool launches. The following screen opens. Touch [Upload].



- 4 Touch [Start] to begin saving the GP project as backup data (*.cml). If there is a password for transferring, input the password first, then touch [Start].



NOTE

- Backup data is saved as displayed in the [File Name] field. If you are using AGP-3500T, by default this is "MEM3X00.CML." If you want to use another file name, touch [New File]. To overwrite an existing file name in the CF Card, touch [Select File].

- 5 Once upload is complete, touch [Back] then [Exit] to close the Memory Loader Tool.
6 Remove the CF Card from the GP and insert it into the PC.